ECON 490: *Capstone Proposal Activity*

In this activity, we'll practice brainstorming capstone project ideas. One of the key skills in developing a successful project is being able to generate *multiple* potential ideas and evaluating them. The goal is to assess whether each idea is viable (i.e., is relevant data available?) and pick the idea that is most promising.

To get started, carefully review the Capstone Data Resources page. Then, do the following:

1) Identify a data set or link that describes a topic in which you're interested (for this activity, try to find data sets/links that are directly available for download).
2) Define a correlation/relationship that is potentially interesting using the variables/columns included in the data set you've selected.
3) Try to apply the idea of "exploring variation" to think about what differences might be interesting in the correlation you've identified (i.e., Does the relationship depend on some other characteristic or variable in the data set? Does it vary across time/states/types of people?).

After going through the steps above, respond to the following questions:

1) What data set did you decide to use? What made you interested in that data set?

2) What correlation/relationship would you estimate with this data? Why is this interesting/what might it tell us?

NOTE: *This answer should be very specific, meaning you should reference specific columns/variables from the data set or documentation.*

3) How would you apply the idea of exploring variation to expand the correlation analysis in question 2? What other characteristics might mediate or impact the correlation you estimate in question 2?

HINT: *Think about other factors that might matter for the basic relationship you identified above—might the relationship change over time? Might it be different in different places? Are there potential omitted variables that you might want to include to get a better estimate of the "true" relationship?*

4) Use the AI prompt posted on GitHub to evaluate your idea. Email me a link for your conversation thread (most AI platforms have an option to share conversations). Ideally, using a "Pro"-level model with reasoning capability (i.e., use Thinking mode in ChatGPT/Gemini). In the space below, briefly describe the quality of feedback you received. Was it helpful?